

HUY M. NGUYEN

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Education

VNU-HCM High School For The Gifted

HCMC

Grade 9 GPA: 9.5 (3.8) (Salutatorian)

Sep 2024 - June 2027

Grade 10 GPA: 9.3 (3.72)

Grade 11 GPA: 9.5 (3.8)

Standardized Tests

IELTS: 7.0

Oct 2024

SAT: 1380

Dec 2025

Honors & Awards

Gold Medal | AI-JAM US 2025

Aug 2025

Developed "SmartGuard Wrist," an AI-powered wearable for real-time disease prevention and emergency risk management.

Gold Medal | iWISE Global Final 2026

Feb 2026

Engineered an AI-based system for automated monitoring of water quality and shrimp health in aquaculture environments.

Physics Brawl (International Physics Competition)

2024

Earned a Certificate of Participation for successfully competing in a high-intensity, global-scale physics problem-solving challenge.

Olympic OPHO (International Physics Olympiad)

2024

Recognized with a Certificate of Participation for demonstrating advanced theoretical physics knowledge at the international level.

First Prize | Excellent Student in Physics - Grade 9 City Level

2023 - 2024

Awarded by Ho Chi Minh City Department of Education and Training for outstanding achievement in Physics.

Outstanding Student Award | Saigon Practical High School

2023 - 2024

Recognized as "Excellent Student" by Saigon Practical High School (affiliated with Saigon University).

Research Experience

Title: Improvements of Performance of Diabetes Prediction Using Combined Machine Learning

Oct - Dec 2025

Models and Explainable AI Techniques - [\[Link\]](#)

Published in the IJMSM, Vol. 2, Issue 4, 2025

- Advanced Data Preprocessing:** Integrated public (Pima Indian) and private datasets, implementing SMOTE and ADASYN algorithms to effectively handle class imbalance and missing data.
- Algorithmic Optimization:** Evaluated multiple classifiers (SVM, XGBoost, Random Forest) using GridSearchCV for hyperparameter tuning, achieving a peak prediction accuracy of 83% through ensemble learning.
- Explainable AI Implementation:** Leveraged SHAP and LIME frameworks to interpret model decisions, identifying Glucose, BMI, and Age as the most critical features for predictive reliability.

Personal Project

Product name: IoT-Based Smart Locker System with Mobile Application Control

2025

- System Integration & Architecture:** Built an IoT-based smart locker system integrating ESP32 hardware, sensors, and electronic lock module with Flutter App, Flask Backend, and Firebase.
- Security & Remote Management:** Enabled secure remote control and multi-factor authentication (OTP, RFID, biometrics) with real-time monitoring and push notifications.
- Optimization & Scalability:** Achieved core features: remote access, safe authentication, instant status updates, and user-friendly interface; future directions include AI anomaly detection, MQTT optimization, and multi-device management.

Leadership Experience

Vice Head of Academic Department (Promoted from Member) – The Promethean Flame - Managed a team of nearly 20 members to orchestrate community tutoring programs and internal operations. - Designed and taught Math and Physics curricula directly to 50+ underprivileged children.	2024 - Present
Head of Chapter HCMC – ADHD Kids: Embrace & Empower (AKEE) - Led local operations, fundraising, and media campaigns, advocating for ADHD awareness to 100+ families and schools. - Orchestrated 50+ workshops and events, driving community engagement and expanding local partnerships.	2025 - Present
Event MC & Co-Host – Lucky Village Event - Smiling Chances - Served as the Lead MC and Co-host for the main stage, engaging a crowd of 80+ children through interactive storytelling and public speaking. - Orchestrated event logistics and co-designed interactive games, ensuring seamless execution and dynamic crowd control. - Extended operational support to healthcare charity drives, coordinating the distribution of medical care packages at Cho Ray Hospital.	Jul 2025
Founder – Heart2Heal (Non-profit Project) - Led a 50-member team to launch digital media campaigns, raising public awareness on community health. - Secured 10,000,000+ VND in donations through online fundraising initiatives to support medical and essential needs. - Directed the distribution of relief packages, directly impacting over 100 underprivileged children.	2025 - Present

Socials Work & Activities

Content Member – Project Alpha (Non-profit Initiative) - Created engaging promotional content for a non-profit initiative, helping secure 7,000,000 VND in donations. - Coordinated with 20 volunteers to execute charity drives and deliver essential gifts to students in need.	2024 - 2025
Core Content Member – Trạng Chuyển Cấp 26:29 - Authored 10+ academic blogs on high school transition strategies, providing critical guidance for incoming students. - Reviewed and peer-edited educational content within the department to ensure rigorous academic standards.	2025 - Present
Volunteer Member – Kindness Bookshelf Initiative - Tủ Sách Nhân Ái - Managed inventory and packaged 3,000+ academic books and educational materials for bi-annual shipments to remote highland schools. - Streamlined sorting and logistics processes within the volunteer team to ensure safe and timely delivery to over 500 underprivileged students.	2022 - Present

Skills & Interests

Technical Skills: C/C++, Python, Machine Learning, IoT Architecture (ESP32, Arduino), Circuit Design, 3D Printing.

Management Skills: Project Management, Team Leadership, Event Logistics, and Budget Allocation.

Languages: Vietnamese (Native), English.

Interests & Talents: Classical Piano, Competitive Football & Badminton.